§3 The three principles: SM, EU and FR — part 2 of 5

Reference: subsec:viab_core, subsec:env_coupling

Source: paper/section3_principles.tex, lines 991–2359 (§3.4 the E_{req} robust viability kernel and the finite-step robust feasible sets; §3.5 the structural environment coupling condition with its auxiliary descriptions and diagnostic quantities), at commit e6e12cd. The Japanese manuscript is authoritative. Part 2 of §3 follows 03_principles_1.md. Part 3 of §3 follows in 03_principles_3.md.

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§3.4 The E_{req} robust viability kernel and the finite-step robust feasible sets

Reference: subsec:viab_core

This subsection introduces the set of states from which a single common action keeps feasibility for every external environment state in $E_{\mathrm{req}}(x)$, together with its finite-step version and a step count (classification C). They describe the future-directed side of the principle response item of SM. They are not truth conditions of any time-indexed satisfaction predicate. subsec:fr and subsec:sd_necessity use the results.

Required-environment-satisfying state domain and the kernel

Definition (def:req_domain): required-environment-satisfying state domain. $D_{\mathrm{req}} := \{x \in X \mid E_{\mathrm{req}}(x) \neq \emptyset\}$ and $K_{\mathrm{req}} := K \cap D_{\mathrm{req}}$. Both are derived sets determined uniquely by E and K in the formal system of §2; they are not objects an analyst chooses. Since $E_{\mathrm{req}}(x) \neq \emptyset$ gives some e with $\eta(x,e) = 1$ for all $\eta \in E$, $D_{\mathrm{req}} \subseteq \bigcap_{\eta \in E} D_\eta$ (def:eta_satisfiable_domain). §2 does not require $E_{\mathrm{req}}(x) \neq \emptyset$ for all x: (CR-1b) requires only the existence of such an x. Hence $D_{\mathrm{req}} \subsetneq X$ and $K_{\mathrm{req}} \subsetneq K$ are not excluded.

Definition (def:viab_core): E_{req} robust viability kernel. Under $x_{t+1} = f(x_t, u_t, e_t)$, a subset $K' \subseteq K_{\mathrm{req}}$ is **E_{req}-invariant** if

$$\forall x \in K',\ \exists u \in U \text{ s.t. } \forall e \in E_{\mathrm{req}}(x),\ f(x,u,e) \in K'. \tag{$*$}$$

Let $\mathfrak{K}$ be the family of all E_{req}-invariant subsets of K_{req}. The E_{req} **robust viability kernel** is $\mathrm{Viab}_f(K) := \bigcup_{K' \in \mathfrak{K}} K'$, and $\mathrm{Viab}_f(K) \subseteq K_{\mathrm{req}} \subseteq K$.

- *Quantifier order.* $(*)$ has the order $\exists u\, \forall e$: **one common action** must preserve feasibility against **all** external environment states in $E_{\mathrm{req}}(x)$. This differs from the condition $\forall e \in E_{\mathrm{req}}(x),\ \exists u :\ f(x,u,e) \in K'$, which lets the action depend on the environment state. The paper does not interchange the two orders.
- *Restriction of the domain.* Elements of $\mathfrak{K}$ are restricted to subsets of K_{req} so that a state with $E_{\mathrm{req}}(x) = \emptyset$ (where the universal quantifier holds vacuously) is not admitted into the kernel. States outside K_{req} are outside the scope of this diagnosis; this paper does not state, as a general consequence, that such a state fails in the future.
- *Relation to the standard viability kernel.* The definition is a construction specific to this paper, using a robust condition over $E_{\mathrm{req}}(x)$. Mathematical identity with the standard viability kernel is not claimed.

Proposition (prop:viab_core_max): existence of the maximal E_{req}-invariant set. $\mathrm{Viab}_f(K)$ is E_{req}-invariant and is therefore the greatest element of $\mathfrak{K}$. *Proof idea:* $\emptyset \in \mathfrak{K}$, so the union is defined (it is $\emptyset$ if $\mathfrak{K} = \{\emptyset\}$). For x in the union, take a $K' \in \mathfrak{K}$ containing x and the common action u given by invariance of K'; since $K' \subseteq \mathrm{Viab}_f(K)$, the same u sends all $e \in E_{\mathrm{req}}(x)$ into $\mathrm{Viab}_f(K)$. No assumption is used.

Structural relation to Viability Theory (`rem:viability_relation`). The robust invariant-set construction relative to E_{req} is structurally related to the ideas of feasibility and invariance in Viability Theory (Aubin 1991; Saint-Pierre 1994) and set-valued analysis (Aubin and Frankowska 1990). The paper uses compactness (`ass:compact_spaces`) and continuity of f (`ass:f_continuity`) elsewhere, but in `def:viab_core` it adopts the paper-specific condition $\exists u\, \forall e \in E_{\text{req}}(x)$ with a state-dependent quantification domain. Identity with the standard viability kernel, or conformity with a particular standard axiom system, is not claimed; the relation is a conceptual and structural correspondence about keeping states inside a feasible set. Closed-graph, upper-semicontinuity or compact-valuedness of $E_{\text{req}} : X \rightrightarrows E_{\text{ext}}$ are not assumed in §2 in general (`def:acs`(3) imposes no regularity on η). The paper therefore does not claim from general Viability Theory that $\text{Viab}_f(K)$ is closed, and uses no claim that requires closedness in this subsection.

Status of the kernel

Ontological and epistemic status (`rem:viab_core_status`). $\text{Viab}_f(K)$ is not a cognitive representation implemented inside the ACS, not something the ACS computes, and not a set the ACS directly aims at. It is a model-derived analysis object that characterizes from which states E_{req}-robust feasibility retention is possible once f, K, U and E_{req} are fixed. Teleological or cognitive expressions ("the ACS aims at the kernel", "tries to return to it", "recognizes it") are not used. *Model relativity:* the kernel is not chosen arbitrarily — it is mathematically determined once the model is fixed — but which f, K and required environmental factors are adopted for a real ACS involves modelling. The paper therefore does not treat an observer-independent, known kernel as given for reality; it treats the kernel as an analysis object relative to a specified formal model.

Formal-model kernel, internal prediction and realized trajectory (`rem:three_layers`). Three things are distinguished and not identified: (1) the kernel on the formal model, derived by the analyst; (2) the ACS's internal prediction and evaluation, which use the internal model M, the internal environment estimate $\hat{e}_t$ (`def:internal_env_estimate`) and the mechanism of §4 and are not assumed to agree with the formal model; (3) the realized trajectory $x_0, x_1, x_2, \ldots$. The paper does not assume that a real ACS can grasp the true transition law or future environment states, identify the required factors, compute the kernel, or predict future trajectories. Being able to define $\text{Viab}_f(K)$ in Phase 1 is a separate matter from a real ACS recognizing or computing it. An ACS that does not know the kernel may nonetheless remain inside it for a long time; conversely, favourable internal evaluation with an incomplete internal model does not preclude leaving the formal-model feasible region. What the paper says of real ACSs reaches only the level that local evaluation, response and update can result in remaining in the kernel. Probabilistic, partial-observation and estimation-error models and probabilistic kernels are Phase 2 (L1, within the present framework).

Meaning of being outside the kernel (`rem:viab_core_exterior`). This paper does not state, as a general consequence, that a state outside $\text{Viab}_f(K)$ will certainly fail in the future. $x \notin \text{Viab}_f(K)$ implies only that, under the quantifier condition of `def:viab_core`, there is no common policy giving E_{req}-robust feasibility retention. The realized sequence of environment states need not sweep through all of $E_{\text{req}}(x_s)$, and the ACS may change its internal model and response policy through Update, so a realized trajectory that keeps feasibility is not excluded. "Being outside the kernel" is not described as "future failure is settled". *Relation to inter-time change:* the kernel is an analysis object for the fixed structure $(f, K, U, E_{\text{req}})$ of Phase 1, whereas §5 treats change of internal model, memory and response policy through Update; the paper does not say that an ACS that has left the kernel does not recover (see `rem:s5_learning_viability` in §5).

Finite-step sets and the step count (classification C)

Definition (`def:viab_stage`): ν-step E_{req} robust feasible set (classification C). With $K_{\text{req}} = K \cap D_{\text{req}}$: $\text{Viab}_f^{(0)}(K) := K_{\text{req}}$, and for $\nu \geq 1$,

$$\text{Viab}_f^{(\nu)}(K) := \{x \in K_{\text{req}} \mid \exists u \in U\ \forall e \in E_{\text{req}}(x) :\ f(x,u,e) \in \text{Viab}_f^{(\nu-1)}(K)\}.$$

The superscript (ν) denotes the number of iteration steps. The quantifier order is $\exists u\, \forall e$, as in `def:viab_core`; elements are restricted to K_{req}, so vacuous cases with $E_{\mathrm{req}}(x) = \emptyset$ are excluded.

Definition (`def:viab_horizon`): robust persistence horizon (classification C). For $x \in K_{\mathrm{req}}$, $\mathrm{Hor}_f(x) := \sup\{\nu \in \mathbb{N}_0 \mid x \in \mathrm{Viab}_f^{(\nu)}(K)\} \in \mathbb{N}_0 \cup \{\infty\}$. The set is non-empty because $x \in \mathrm{Viab}_f^{(0)}(K)$.

Proposition (`prop:viab_stage_nested`): nesting. (i) $\mathrm{Viab}_f^{(\nu+1)}(K) \subseteq \mathrm{Viab}_f^{(\nu)}(K)$ for all ν. (ii) For $x \in K_{\mathrm{req}}$: $x \in \mathrm{Viab}_f^{(\nu)}(K) \iff \nu \le \mathrm{Hor}_f(x)$; if $\mathrm{Hor}_f(x) < \infty$ the supremum is attained as a maximum. *Proof idea:* induction for (i); (ii) follows from downward closedness of the index set.

Proposition (`prop:viab_stage_core`): kernel inside every finite-step set. $\mathrm{Viab}_f(K) \subseteq \mathrm{Viab}_f^{(\nu)}(K)$ for all ν; hence $x \in \mathrm{Viab}_f(K) \Rightarrow \mathrm{Hor}_f(x) = \infty$. *Proof idea:* induction, using E_{req}-invariance of $\mathrm{Viab}_f(K)$ (`prop:viab_core_max`).

Proposition (`prop:viab_stage_finite`): finite state space. If X is finite, there is $\nu^* \le |K_{\mathrm{req}}|$ with $\mathrm{Viab}_f^{(\nu)}(K) = \mathrm{Viab}_f^{(\nu^*)}(K)$ for all $\nu \ge \nu^*$, and

$$\mathrm{Viab}_f(K) = \mathrm{Viab}_f^{(\nu^*)}(K) = \bigcap_{\nu \in \mathbb{N}_0} \mathrm{Viab}_f^{(\nu)}(K).$$

Hence, for $x \in K_{\mathrm{req}}$, $\mathrm{Hor}_f(x) = \infty \iff x \in \mathrm{Viab}_f(K)$. The proposition uses no assumption. *Proof idea:* the sequence is decreasing in a finite set, so it stabilizes; with $\Phi(V)$ the one-step operator, $V^{(\nu+1)} = V^{(\nu)}$ implies all later terms coincide; the stable set is E_{req}-invariant, so it lies in the kernel, and the reverse inclusion is `prop:viab_stage_core`.

Status and limits of the step count (`rem:viab_stage_scope`).

1. This paper does not claim $\mathrm{Viab}_f(K) = \bigcap_\nu \mathrm{Viab}_f^{(\nu)}(K)$ in general. Elements of the right-hand side are robust at each finite step, but the paper has not shown that the right-hand side is E_{req}-invariant. For the same reason it does not claim $\mathrm{Hor}_f(x) = \infty \Rightarrow x \in \mathrm{Viab}_f(K)$ in general. For finite X both hold by `prop:viab_stage_finite`.
2. $\mathrm{Hor}_f(x)$ is a number of steps of robustness over all of $E_{\mathrm{req}}(x)$, not a forecast of the number of time steps survived under the realized environment sequence.
3. Hor_f is not a truth condition of SM_t; it is a classification-C quantity describing the future-directed side of the principle response item of SM.
4. A system can have $\mathrm{Viab}_f(K) = \emptyset$ while $\mathrm{Viab}_f^{(\nu)}(K) \neq \emptyset$ for some ν. Type-level example in the manuscript: $X = \{0,1,2\} \subseteq \mathbb{R}$, $\Pi(x) = x-1$ (so $K = \{1,2\}$), $U = \{u\}$, $E_{\mathrm{ext}} = \{e\}$, $E = \{\eta\}$, $\eta \equiv 1$, $f(2,u,e) = 1$, $f(1,u,e) = f(0,u,e) = 0$; then $\mathrm{Viab}_f^{(1)}(K) = \{2\}$, $\mathrm{Viab}_f^{(2)}(K) = \emptyset$ and, by `prop:viab_stage_finite`, $\mathrm{Viab}_f(K) = \emptyset$. The paper does not claim that finite-step sets are non-empty in every system.
5. The ACS is not required to hold or compute these quantities (`subsec:description_levels_global`).

Comparison with the all-environment robust kernel and the optimistic kernel

For comparison, two kernels are introduced whose quantification ranges over the whole of E_{ext} rather than $E_{\mathrm{req}}(x)$.

Definition (`def:kernel_all_any`): all-environment robust kernel and optimistic kernel (classification C). A subset $K' \subseteq K$ is **all-environment robust invariant** if $\forall x \in K',\ \exists u \in U,\ \forall e \in E_{\mathrm{ext}} :\ f(x,u,e) \in K'$, and **optimistic invariant** if $\forall x \in K',\ \exists u \in U,\ \exists e \in E_{\mathrm{ext}} : f(x,u,e) \in K'$. The unions of all such subsets of K are $\mathrm{Viab}_f^{\mathrm{all}}(K)$ and $\mathrm{Viab}_f^{\mathrm{any}}(K)$. The labels all, any denote the kind of quantification and differ from the parenthesized step superscript (ν). These kernels use neither E nor E_{req}, so unlike $\mathrm{Viab}_f(K)$ they need not lie in K_{req}. They

are comparison sets, not truth conditions of any time-indexed satisfaction predicate.

Proposition (`prop:kernel_standard_comparison`): inclusions. (i) $\mathrm{Viab}_f^{\mathrm{all}}(K)$ is all-environment robust invariant and $\mathrm{Viab}_f^{\mathrm{any}}(K)$ is optimistic invariant. (ii) $\mathrm{Viab}_f(K) \subseteq \mathrm{Viab}_f^{\mathrm{any}}(K)$. (iii) $\mathrm{Viab}_f^{\mathrm{all}}(K) \subseteq \mathrm{Viab}_f(K)$ holds if and only if $\mathrm{Viab}_f^{\mathrm{all}}(K) \subseteq D_{\mathrm{req}}$. No assumption is used. *Proof idea:* (ii) uses $E_{\mathrm{req}}(x) \neq \emptyset$ on the kernel to pick one e; (iii) uses $E_{\mathrm{req}}(x) \subseteq E_{\mathrm{ext}}$, so all-environment robustness implies E_{req}-robustness once the set lies in K_{req}.

Reading and limits (`rem:kernel_standard_comparison_scope`). (1) By (iii), a state robust against all external environment states but lying outside D_{req} is not in the kernel: the kernel expresses robustness over the range where the required factors are satisfied, not over all of E_{ext}. (2) The paper does not state, as a general consequence, whether the inclusions in (ii) and (iii) are proper or equalities; this depends on f, K and E. (3) Definitional identity with the differentiated kernel or other existing kernels is not claimed (`subsec:related`). (4) The ACS is not required to hold or compute these sets.

Figure (`fig:s3_kernel_inclusion`). The manuscript's figure shows (a) the inclusions that hold without extra conditions — $\mathrm{Viab}_f(K) \subseteq \mathrm{Viab}_f^{(\nu+1)}(K) \subseteq \mathrm{Viab}_f^{(\nu)}(K) \subseteq \mathrm{Viab}_f^{(0)}(K) = K_{\mathrm{req}} \subseteq K$ and $\subseteq D_{\mathrm{req}}$, $\mathrm{Viab}_f(K) \subseteq \mathrm{Viab}_f^{\mathrm{any}}(K) \subseteq K$, $\mathrm{Viab}_f^{\mathrm{all}}(K) \subseteq K$ — and (b) the conditional relation of `prop:kernel_standard_comparison`(iii) and the finite-X equality of `prop:viab_stage_finite`. Pairs without a line are not shown as included, and the figure does not show whether an inclusion is proper or an equality. The figure itself is not reproduced in this summary.

Robustness that takes the response of the action-response component into account

Under $E_{\mathrm{ext}} = E^{\mathrm{bg}} \times O$ (`def:Eext_decomposition`), the quantification of `def:viab_core` includes, for the action-response component o, all values that could have occurred at the same time. This part instead fixes the current o and lets its next value be set by the transition map g_O (`def:env_realization`).

Definition (`def:viab_resp`): response-inclusive robust viability kernel (classification C). For $(x, o) \in X \times O$, the **background-satisfaction set** is $B_{\mathrm{req}}(x, o) := \{b \in E^{\mathrm{bg}} \mid (b, o) \in E_{\mathrm{req}}(x)\}$, and $K_{\mathrm{req}}^{\mathrm{resp}} := \{(x, o) \in K \times O \mid B_{\mathrm{req}}(x, o) \neq \emptyset\}$. A subset $K' \subseteq K_{\mathrm{req}}^{\mathrm{resp}}$ is **response-inclusively E_{req}-invariant** if

$$\forall (x, o) \in K',\ \exists u \in U \text{ s.t. } \forall b \in B_{\mathrm{req}}(x, o),\ \big(f(x, u, (b, o)),\ g_O(o, x, u, b)\big) \in K'.$$

With $\mathfrak{K}^{\mathrm{resp}}$ the family of such subsets, $\mathrm{Viab}_f^{\mathrm{resp}}(K) := \bigcup_{K' \in \mathfrak{K}^{\mathrm{resp}}} K' \subseteq X \times O$. Requiring $B_{\mathrm{req}}(x, o) \neq \emptyset$ excludes vacuous cases, as K_{req} does in `def:viab_core`. The quantifier order is again $\exists u\, \forall b$.

Proposition (`prop:viab_resp_max`). $\mathrm{Viab}_f^{\mathrm{resp}}(K)$ is response-inclusively E_{req}-invariant and is the greatest element of $\mathfrak{K}^{\mathrm{resp}}$ (same argument as `prop:viab_core_max`).

Proposition (`prop:robust_noninclusion`): non-inclusion of the two robustness notions. Write $\pi_X(\mathrm{Viab}_f^{\mathrm{resp}}(K)) = \{x \mid \exists o :\ (x, o) \in \mathrm{Viab}_f^{\mathrm{resp}}(K)\}$. For tuples $(X, U, E^{\mathrm{bg}}, O, f, K, E, g_O)$ of the type to which both definitions apply, neither inclusion between $\mathrm{Viab}_f(K)$ and $\pi_X(\mathrm{Viab}_f^{\mathrm{resp}}(K))$ holds in general: examples exist for (i) $\mathrm{Viab}_f(K) \not\subseteq \pi_X(\mathrm{Viab}_f^{\mathrm{resp}}(K))$ and (ii) $\pi_X(\mathrm{Viab}_f^{\mathrm{resp}}(K)) \not\subseteq \mathrm{Viab}_f(K)$. Both examples (in the manuscript) use $X = \{0, 1\}$, $\Pi(x) = x - 1$ (so $K = \{1\}$), $U = \{u_\emptyset\}$, one frame, $E = \{\eta\}$.

- In (i), taking the action-response component to food (o = food) leaves $g_O \equiv \bot_a$: after intake the component is lost, so the response-inclusive kernel is empty while $1 \in \mathrm{Viab}_f(K)$.
- In (ii), the component stays at "good" under the ACS ($g_O \equiv$ good), so $\{(1, \text{good})\}$ is response-inclusively invariant, while $\mathrm{Viab}_f(K) = \emptyset$ because a value $(b_1, \text{bad}) \in E_{\mathrm{req}}(1)$ that does not actually occur is still quantified over.

Reading and scope of the two examples (`rem:robust_noninclusion_reading`). The examples are built at the level of type, for which the definitions apply. The proposition does not claim that they are individuals satisfying all constitutive requirements and assumptions of §2.

Quantification over alternatives at the same time (`rem:same_time_alternatives`). In §3, conditions quantified over E_{ext} or $E_{\text{req}}(x)$ — the one-step critical domain $\text{Crit}_{\eta}^{(1)}$, $\mathcal{D}_{\text{FR}}$, μ_{eff}, $\text{EC}(x)$, the robustness of `def:viab_core` and `def:viab_stage`, and the response-incapable environment set of `subsec:sd_necessity` — include, also for the action-response component, other values possible at the same time, without asking whether a value is reachable from the history. Their truth conditions do not change across the decomposition of §2. Only `def:viab_resp` has a response-inclusive version; response-inclusive versions of the response-incapable environment set, μ_{eff} and the conditional analysis of `subsec:sd_necessity` are tasks for Phase 1.α (`subsec:phase1a`).

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§3.5 The structural environment coupling condition and its auxiliary descriptions and diagnostic quantities

`Reference: subsec:env_coupling`

This subsection formalizes, independently of the three principles, the structural coupling by which the ACS is not functionally closed against differences of external environment states, and positions it as a premise and background structure of EU. It also organizes the environment-response component, the sensitivity of the generalized gradient in the environment-response directions, and the realized external interaction as auxiliary descriptions and diagnostic quantities. "Auxiliary" means they are not truth conditions of the time-indexed satisfaction predicates of the principles.

Three things are not identified:

1. **Structural environment coupling:** whether a structure exists by which differences of external environment states can be reflected in the ACS's next state (`def:env_coupling`).
2. **Realized environment interaction / environment sensitivity:** whether a boundary-crossing external action is realized (`def:phi`), and whether the component determining the feasibility margin belongs to an environment-response coordinate (`prop:pi_E_sensitivity`).
3. **EU:** whether all one-step critical required factors are satisfied (`def:EU`).

The environment-response component x_E

Assumption (`ass:env_response`): structural determination of the environment-response component. For the ACS Σ a coordinate system on X is given. Relative to the formal description of the ACS, an index set $I_E \subseteq \{1, \dots, n\}$ with $|I_E| = n_E$, $1 \le n_E \le n$, of the components related to responses to differences of external environment states is fixed. In Phase 1, I_E is treated as a structural parameter specific to the ACS, not as an evaluation set chosen arbitrarily by the analyst for judging the coupling condition. The assumption does not require any particular neural circuit, algorithm, sensor configuration or hardware. *Introduction status: S (structural). Strength: medium.* Relied on by `def:x_E` and `prop:pi_E_sensitivity`.

Definition (`def:x_E`): environment-response component x_E and coordinate projection π_E. With $I_E = \{k_1 < \cdots < k_{n_E}\}$, $\pi_E : \mathbb{R}^n \to \mathbb{R}^{n_E}$, $\pi_E(z) := (z_{k_1}, \dots, z_{k_{n_E}})$. It is linear, and is applied both to $x_t \in X \subseteq \mathbb{R}^n$ and to generalized-gradient vectors $g \in \partial\Pi(x_t) \subseteq \mathbb{R}^n$; its domain is taken to be $\mathbb{R}^n$ to avoid type ambiguity. $x_E := \pi_E(x_t) \in \mathbb{R}^{n_E}$ is the **environment-response component**, an internal state component of $x_t \in X$; it does not denote the realized external environment state e_t, the space E_{ext}, or the required environment space E. $\pi_E(X)$ is compact (compactness of X and continuity of π_E). The **projected generalized-gradient set** is $\partial_{x_E}\Pi(x_t) := \{\pi_E(g) \mid g \in \partial\Pi(x_t)\}$, a non-empty compact convex set (linear image of $\partial\Pi(x_t)$; `prop:pi_subdiff`). The condition $\partial_{x_E}\Pi(x_t) \neq \{\mathbf{0}_{n_E}\}$ is an auxiliary condition on environment sensitivity related to structural coupling, but it is not the structural environment coupling condition (`def:env_coupling`) itself.

Observational correspondence (`rem:env_response_observation`). In real cognitive systems, a component that responds systematically differently to different external environment states e, e' can be an observational candidate for I_E; formally, a component k is a candidate when $[f(x,u,e)]_k \neq [f(x,u,e')]_k$ is confirmed for some x, u and $e \neq e'$. This is not a general causal identification rule for I_E; operational and causal criteria that take observation noise, delays and internal feedback into account are Phase 2.

The structural environment coupling condition

Definition (`def:env_coupling`): structural environment coupling condition (classification C). A state $x \in X$ satisfies the condition, written $\mathrm{EC}(x)$, if

$$\exists u \in U,\ \exists e, e' \in E_{\mathrm{ext}} \text{ s.t. } e \neq e',\ \pi_E(f(x,u,e)) \neq \pi_E(f(x,u,e')).$$

For time t one uses $\mathrm{EC}(x_t)$. It is a minimal Phase 1 structural condition expressing that an ACS with a self–environment boundary is not functionally closed against differences of external environment states; it is not a time-indexed satisfaction predicate of a principle. It formalizes only part of the input-side structure. An independent environment dynamics such as a generation law for the exogenous background sequence is not formulated in Phase 1; the transition of the action-response component is given only as the map g_O of `def:env_realization` (L2 in `subsec:limitations_list`). The definition does not include as a defining condition that a relation to a particular external object is currently realized, that the external state is actually reachable, that required environmental factors are satisfied, that the time-indexed predicate of EU is true, or that relation quality is positive.

Coupling versus generalized-gradient sensitivity (`rem:ro_two_layer_intuition`). $\mathrm{EC}(x)$ describes a *structural coupling possibility*: differences of external environment states can non-trivially affect the next state of the environment-response component. By contrast, the condition of `prop:pi_E_sensitivity`, $\exists g \in \partial\Pi(x_t) :\ \pi_E(g) \neq \mathbf{0}_{n_E}$, is a *local feasibility-sensitivity condition*: at least one of the weakest components determining the current feasibility margin belongs to an environment-response coordinate. They are different conditions, and the paper does not state an implication in either direction as a general consequence. Structural coupling possibility, local feasibility sensitivity, realized external interaction, EU and relation quality are distinguished from one another.

Output-side structural cue (`rem:u_ext_output_cue`). The set $U_{\mathrm{ext}} \subseteq U$ of actions that act directly on the external environment (`rem:action_space_structure`, §2) is an output-side structural cue of action from the ACS to the environment. Non-emptiness of U_{ext} alone does not yield, as a general consequence, a realized environment interaction, the existence of critical required factors, or truth of the time-indexed predicate of EU. U_{ext} is not a condition of any time-indexed satisfaction predicate.

The relation between π_E and Π

Proposition (`prop:pi_E_sensitivity`): minimal sensitivity of π_E, equivalence form. Under `ass:env_response` and `prop:pi_subdiff`,

$$[\exists g \in \partial\Pi(x_t) :\ \pi_E(g) \neq \mathbf{0}_{n_E}] \iff I_E \cap \arg\min_k (x_{k,t} - \theta_k) \neq \emptyset.$$

Proof idea: by `prop:pi_subdiff`(ii), $\partial\Pi(x_t) = \mathrm{conv}\{e_k \mid k \in \arg\min_j (x_{j,t} - \theta_j)\}$. If I_E misses the argmin, every $\pi_E(e_k)$ vanishes, so $\pi_E(g) = 0$ for all g (contrapositive of $\Rightarrow$). Conversely, for $k^* \in I_E \cap \arg\min$, $g := e_{k^*} \in \partial\Pi(x_t)$ has $\pi_E(g) = e_r^{(n_E)} \neq 0$.

Local sensitivity in x_E directions (`rem:pi_E_local_sensitivity`). If the condition holds and $x_t \in \mathrm{int}\, X$, decreasing one of the minimizing environment-response components slightly decreases $\Pi(x_t)$; the condition is consistent with Π being locally non-constant in at least one x_E direction at interior points. At boundary points the admissible perturbation directions depend on the local shape of X, so general directional analysis is Phase 2.

Threshold violation of the environment-response component and the role of diagnostic indicators

Definition (`def:ro_related_violation`): threshold-violation set of the environment-response component (classification C). $V^-_E(x_t) := I_E \cap V^-(x_t)$. $V^-_E(x_t) \neq \emptyset$ means that at least one environment-response component is below its threshold.

Diagnostic limits of the min-type Π and the role of complementary indicators (`rem:diagnostic_role`). $\Pi(x_t) = \min_k(x_{k,t} - \theta_k)$ represents only the worst component and does not identify several threshold violations: for $(x_{k,t} - \theta_k)_k = (-2, -1, +3)$, $\Pi = -2$ and the violation -1 of the second component is not distinguished by Π alone. $V^-(x_t)$ keeps all violating components and $d^-(x_t)$ the deficiency of each. Roles: Π — feasibility judgment and a scalar for the worst component; $\partial\Pi = \partial^C\widetilde{\Pi}$ — direction of the weakest components; V^- — all threshold-violating components; d^- — each deficiency; V^-_E — threshold violation of environment-response components. These are theoretical diagnostic quantities and do not mean that the ACS's internal sensing and evaluation detect the same quantities. $V^-_E(x_t)$ is in general a different set from $I_E \cap \arg\min_k(x_{k,t} - \theta_k)$: the latter picks the environment-response components among the weakest ones, whereas V^-_E picks those below threshold whether or not they are weakest. An environment-response component below threshold but not weakest is therefore not captured by the $\partial\Pi$-based test of `prop:pi_E_sensitivity`, while V^-_E captures it. These complementary indicators are diagnostic refinements not needed for the time-indexed predicates or the structural propositions of this section, and their dynamical use is Phase 2 (within the present framework). The paper does not state, as a general consequence, EC from $V^-_E(x_t) \neq \emptyset$; V^-_E, the local sensitivity condition of `prop:pi_E_sensitivity` and EC are separate concepts, and no general necessary-and-sufficient relation among them is assumed.

Description of realized external interaction

Definition (`def:phi`): external interaction indicator $\chi_{\mathrm{EI},t}$ (classification C). $\chi_{\mathrm{EI},t} \in \{0, 1\}$ is 1 if at least one boundary-crossing external action is realized at time t, and 0 if none is. A boundary-crossing external action is not limited to input from outside to the ACS: it may include intake from outside, action/discharge/dissipation from inside to outside, and bidirectional actions, as long as it crosses the self–environment boundary. $\chi_{\mathrm{EI},t} = 1$ expresses only that some boundary-crossing action is realized at t; the paper does not state from it, as a general consequence, that required factors are satisfied, that the interaction suffices for feasibility maintenance, or that its relation quality is positive. $\chi_{\mathrm{EI},t} = 0$ is called the **completely isolated state** at t; it describes the absence of realized boundary-crossing action and is not another name for failure of the structural coupling condition, $x_t \notin K$, falsity of the time-indexed predicate of EU, blockage or collapse.

Status of χ_{EI} (`rem:chi_ei_status`). $\chi_{\mathrm{EI},t}$ is a classification-C descriptive and diagnostic quantity, not a new component of the 7-tuple Σ, and a real ACS is not required to hold or compute the same binary variable. Intensity, flow, directional decomposition, duration, aggregation of several interactions and operational measurement are not fixed in Phase 1.

Definition (`def:terminal_isolation`): terminal complete isolation. A realized trajectory is in terminal complete isolation from t_0 if $\chi_{\mathrm{EI},t} = 0$ for all $t \geq t_0$. This is a stronger temporal description than complete isolation at one time. The definition itself does not entail loss of feasibility, blockage or collapse.

Distinctions among coupling, realized interaction, required-environment satisfaction and EU

Structural coupling possibility versus realized interaction (`rem:ro_realized_interaction_distinction`). $\mathrm{EC}(x_t)$ is a *coupling structure* relative to x_t; $\chi_{\mathrm{EI},t}$ is a *realized event*. The Phase 1 general definitions assume neither $\mathrm{EC}(x_t) \Rightarrow \chi_{\mathrm{EI},t} = 1$ nor $\chi_{\mathrm{EI},t} = 1 \Rightarrow \mathrm{EC}(x_t)$.

E, E_{req}, EC and EU are not identical (`rem:ereq_not_ro`). The required environment space E and the set $E_{\mathrm{req}}(x)$ give the background structure of external dependence — persistence of the ACS is not independent of external conditions. EC describes an ACS-side structural coupling. EU is a time-indexed satisfaction predicate about the satisfaction of critical required factors at each time. The four are related but not identical. The paper does not state, as a general

consequence, $\mathrm{EC}(x_t)$ from $E_{\mathrm{req}}(x_t) \neq \emptyset$ alone, $E_{\mathrm{req}}(x_t) \neq \emptyset$ from $\mathrm{EC}(x_t)$, or truth of EU_t from $\mathrm{EC}(x_t)$.

Auxiliary quantities and the truth condition of EU (`rem:env_coupling_not_eu_truth`). The quantities of this subsection are not truth conditions of the time-indexed predicate of EU. The paper does not state, as a general consequence, falsity of EU_t from $\chi_{\mathrm{EI},t} = 0$, or truth of EU_t from the environment-response sensitivity of `prop:pi_E_sensitivity`. The predicate is determined only by satisfaction of the critical required factors of `def:EU`.

Realized interaction versus required-environment satisfaction (`rem:ei_vs_ereq`). $e_t \in E_{\mathrm{req}}(x_t)$ states that, relative to the current state, the realized environment state satisfies all required factors. It is a condition distinct from $\chi_{\mathrm{EI},t} = 1$. The paper assumes neither $e_t \in E_{\mathrm{req}}(x_t) \Rightarrow \chi_{\mathrm{EI},t} = 1$ nor the converse. Likewise $e_t \notin E_{\mathrm{req}}(x_t)$ means that at least one required factor is unsatisfied at (x_t, e_t); the paper does not state complete isolation $\chi_{\mathrm{EI},t} = 0$ from it as a general consequence, and does not derive $e_t \notin E_{\mathrm{req}}(x_t)$ from complete isolation by definition alone.

Temporal limits on realized interaction

Temporary versus terminal isolation (`rem:temporary_terminal_isolation`). $\chi_{\mathrm{EI},t} = 0$ at a single time is distinguished from terminal complete isolation ($\exists t_0\, \forall t \geq t_0 :\ \chi_{\mathrm{EI},t} = 0$). The former is a temporary absence of realized interaction; the latter is a lasting absence after some finite time. Even for the latter, a connection with persistent lack of CR-1-related required factors, persistence outside K, blockage or collapse must be shown separately. No general rule $\chi_{\mathrm{EI},t} = 0 \Rightarrow \mathrm{Col}_{i,t} = 1$ is introduced on the basis of duration alone.

Position of this subsection

The structural environment coupling condition, the realized external interaction, required-environment satisfaction and the time-indexed predicate of EU belong to different levels of description (`rem:conv_ro_layers`) and are not identified. The structural coupling condition is a background structure of the self–environment relation in which satisfaction of required factors is at issue; the next subsection (`subsec:eu`) defines that satisfaction as the time-indexed satisfaction predicate of EU.

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